KAIZEN MATHS

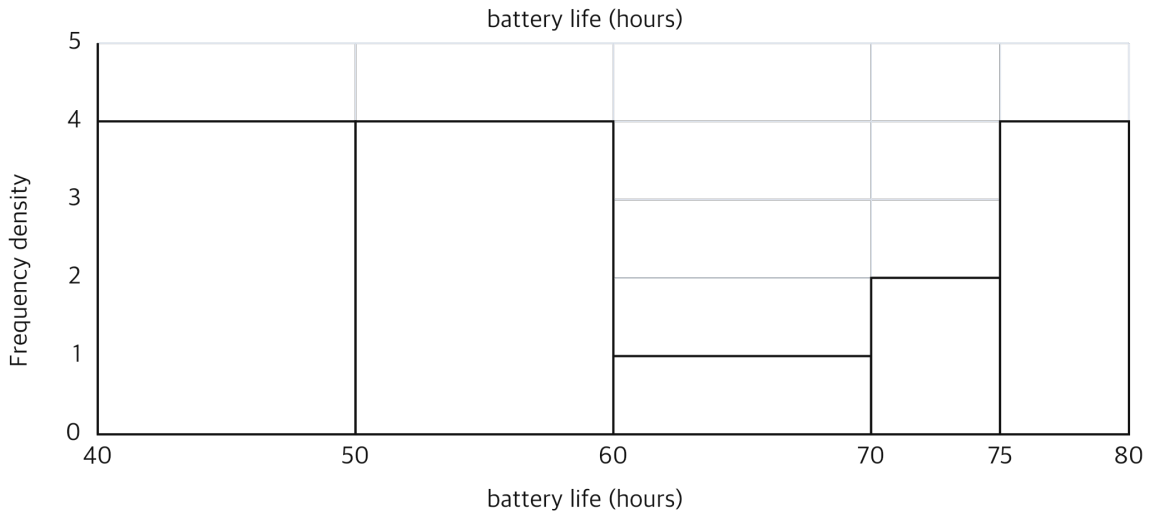
Lesson starter

Histograms and Frequency Density

Interpreting Histograms

Name _____

SA1



State the modal class.

STA-FE00-L2 · SA

Show your working

KAIZEN MATHS

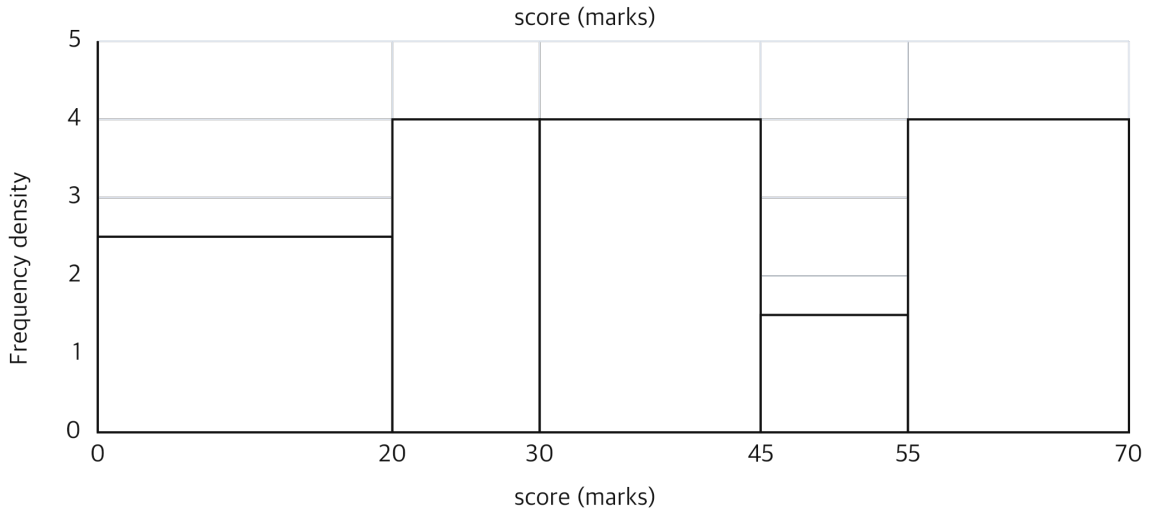
Lesson starter

Histograms and Frequency Density

Interpreting Histograms

Name _____

SB1



Find the frequency in the class $20 \leq x < 30$.

STA-FE00-L2 · SB

Show your working

KAIZEN MATHS

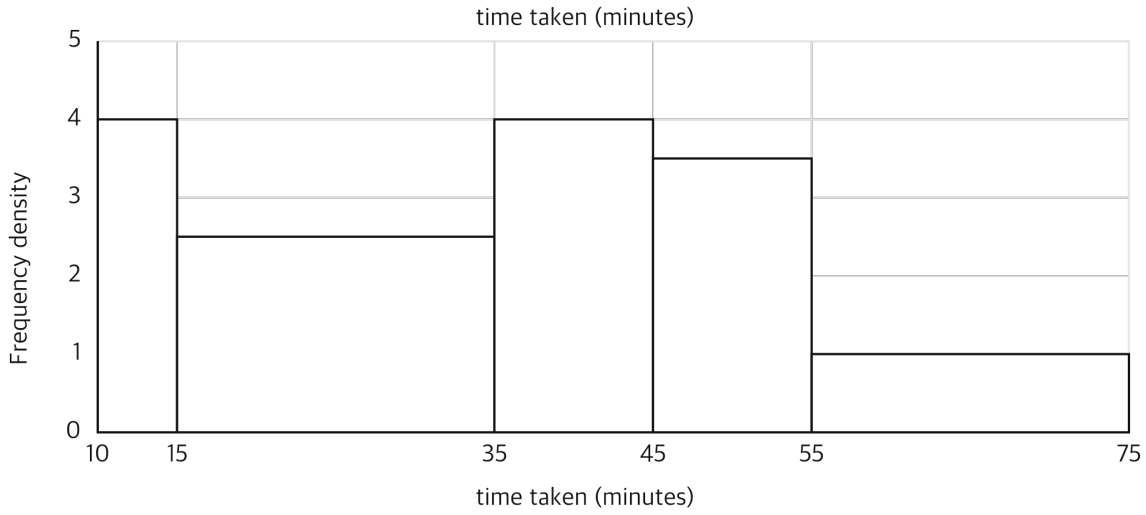
Lesson starter

Histograms and Frequency Density

Interpreting Histograms

Name _____

SC1



Find the total frequency from 15 to 55 minutes.

STA-FE00-L2 · SC

Show your working

KAIZEN MATHS

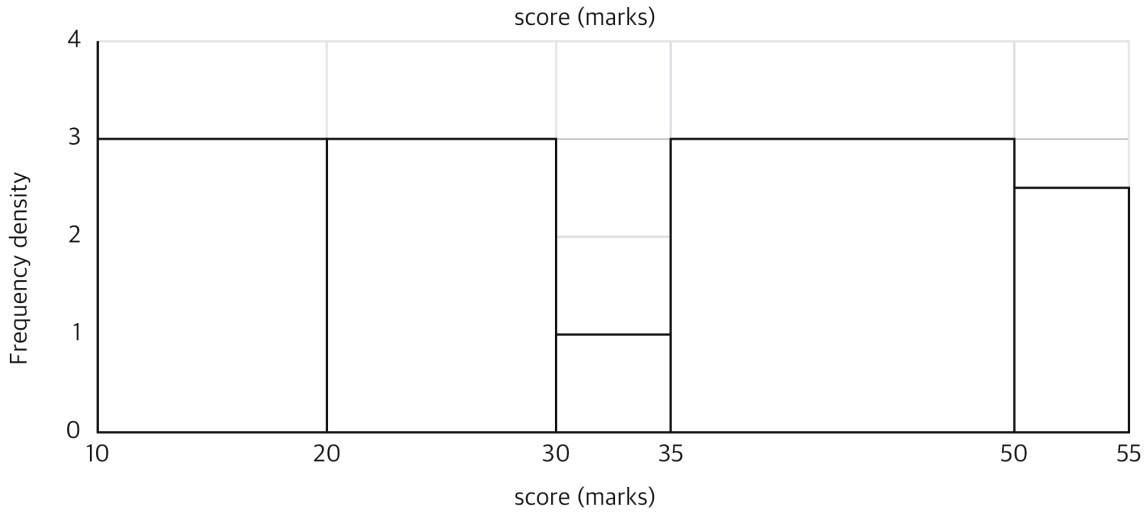
Lesson starter

Histograms and Frequency Density

Interpreting Histograms

Name _____

SD1



Estimate the number of candidates with score from 35 to 42.5 marks.

STA-FE00-L2 · SD

Show your working

KAIZEN MATHS

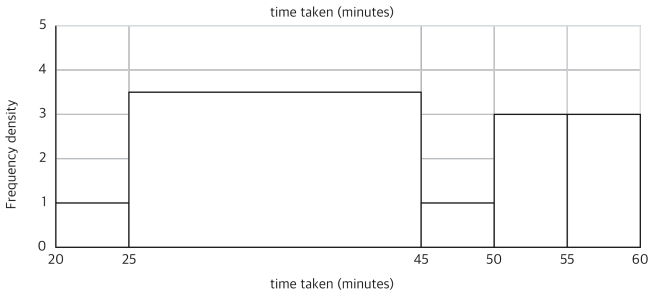
Exit ticket

Histograms and Frequency Density

Interpreting Histograms

Name _____

EA1



Find the percentage of students with time taken from 25 to 55 minutes.

STA-FE00-L2 · EA

Show your working

KAIZEN MATHS

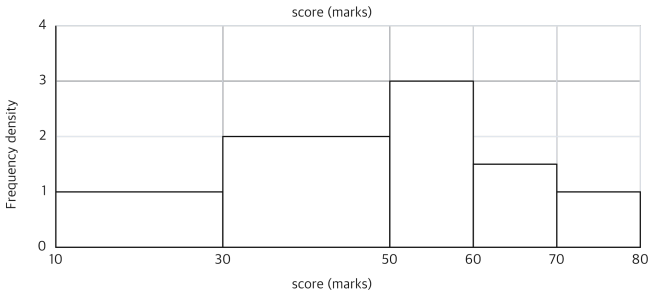
Exit ticket

Histograms and Frequency Density

Interpreting Histograms

Name _____

EB1



State the modal class.

STA-FE00-L2 · EB

Show your working

KAIZEN MATHS

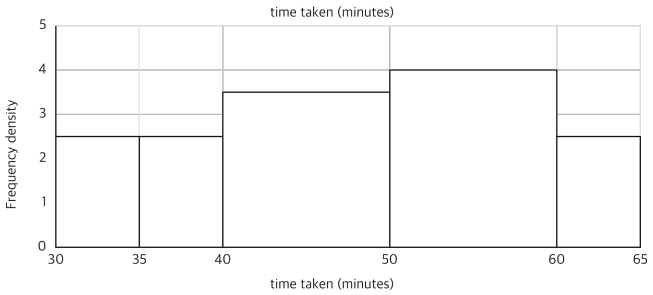
Exit ticket

Histograms and Frequency Density

Interpreting Histograms

Name _____

EC1



Find the frequency in the class $40 \leq x < 50$.

STA-FE00-L2 · EC

Show your working

KAIZEN MATHS

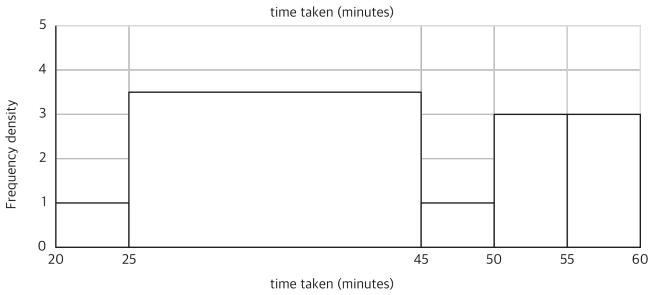
Exit ticket

Histograms and Frequency Density

Interpreting Histograms

Name _____

EA1



Find the percentage of students with time taken from 25 to 55 minutes.

STA-FE00-L2 · EA

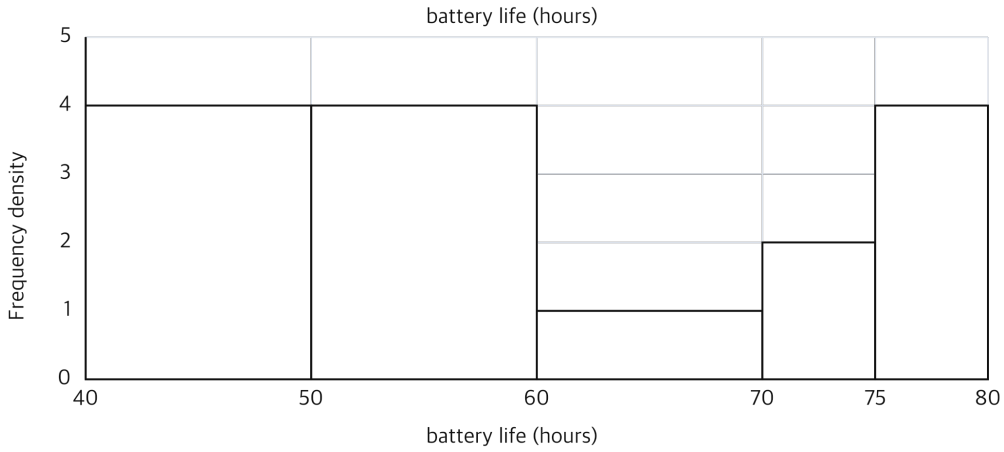
Show your working

Teacher answers & working

Histograms and Frequency Density · Interpreting Histograms

STA-FE00-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



State the modal class.

Answer:

$40 \leq x < 50$

Working:

The modal class is the class with the greatest frequency density.

Look for the tallest bar on the histogram.

The greatest frequency density is 4.

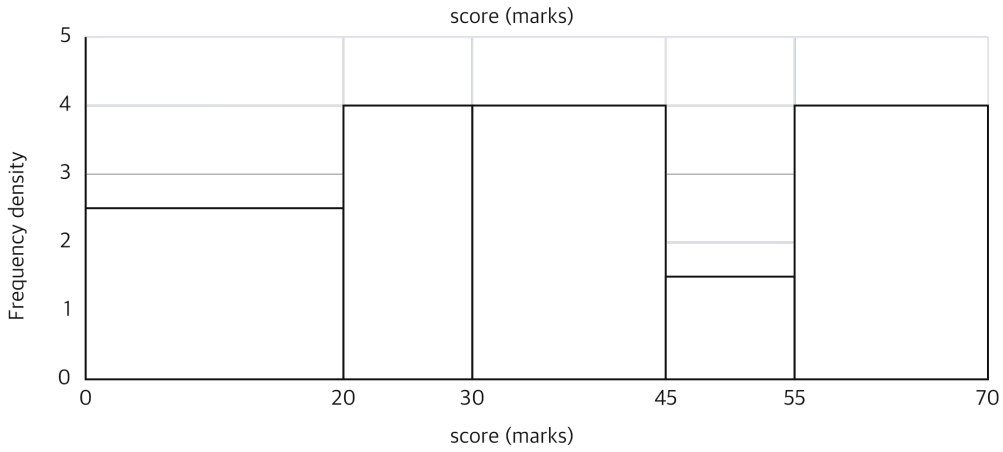
modal class = 40 ≤ x < 50

Teacher answers & working

Histograms and Frequency Density · Interpreting Histograms

STA-FE00-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Find the frequency in the class $20 \leq x < 30$.

Answer:

40 candidates

Working:

In a histogram, frequency is represented by area.

frequency = class width × frequency density

For $20 \leq x < 30$, class width = 30 - 20 = 10.

The height of the bar is the frequency density, 4.

frequency = 10 × 4 = 40

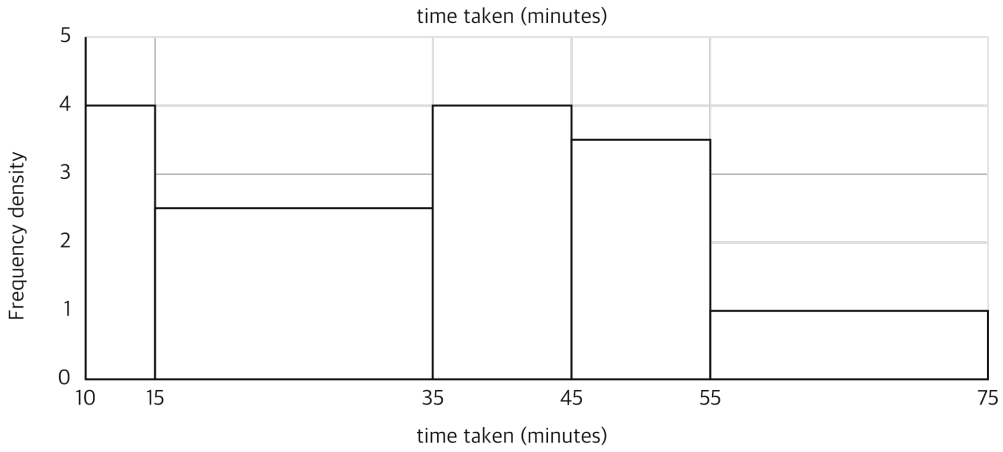
40 candidates

Teacher answers & working

Histograms and Frequency Density · Interpreting Histograms

STA-FE00-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Find the total frequency from 15 to 55 minutes.

Answer:

125 students

Working:

Use the area of each full bar in the interval.

frequency = class width × frequency density

For 15 ≤ x < 35, class width = 35 - 15 = 20.

frequency = 20 × 2.50 = 50

For 35 ≤ x < 45, class width = 45 - 35 = 10.

frequency = 10 × 4 = 40

For 45 ≤ x < 55, class width = 55 - 45 = 10.

frequency = 10 × 3.50 = 35

total frequency = 50 + 40 + 35 = 125

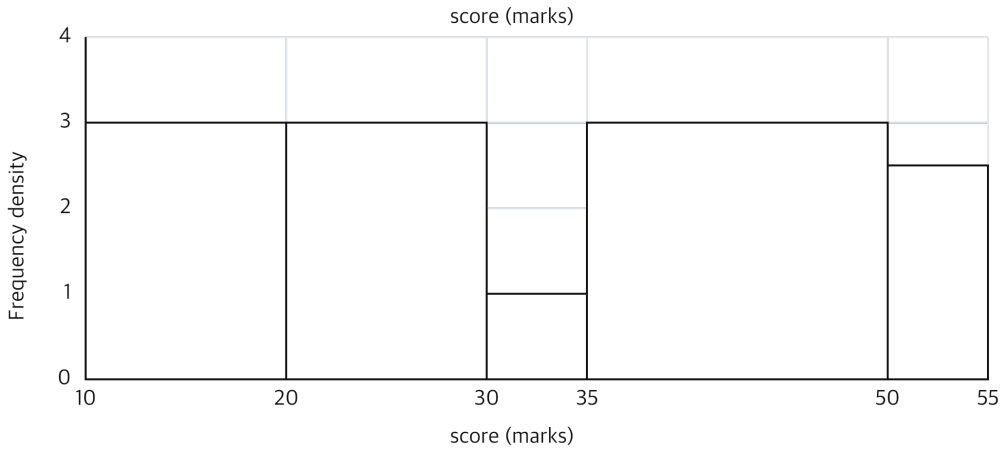
125 students

Teacher answers & working

Histograms and Frequency Density · Interpreting Histograms

STA-FE00-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



Estimate the number of candidates with score from 35 to 42.5 marks.

Answer:

22.50 candidates

Working:

This is a partial interval, so assume the data are evenly spread within the class interval.

estimated frequency = partial width × frequency density

The partial width is 42.5 - 35 = 7.50.

The frequency density for the full class 35 ≤ x < 50 is 3.

estimated frequency = 7.50 × 3 = 22.50

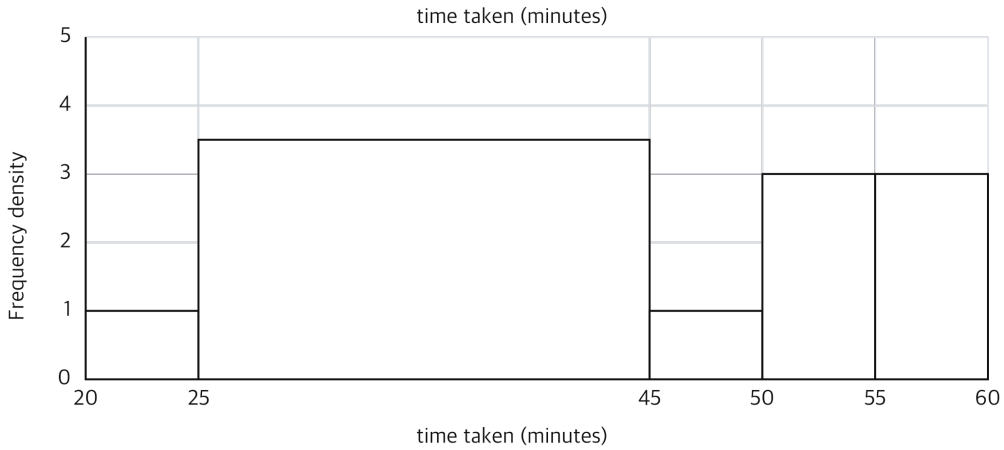
22.50 candidates

Teacher answers & working

Histograms and Frequency Density · Interpreting Histograms

STA-FE00-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



Find the percentage of students with time taken from 25 to 55 minutes.

Answer:

81.8%

Working:

First find the total frequency represented by all bars.

total frequency = 5 + 70 + 5 + 15 + 15 = 110

Now find the frequency in the required interval by adding the relevant bar areas.

$(20 \times 3.50) + (5 \times 1) + (5 \times 3) = 90$

Convert the fraction of the total into a percentage.

percentage = 90 ÷ 110 × 100 = 81.8%

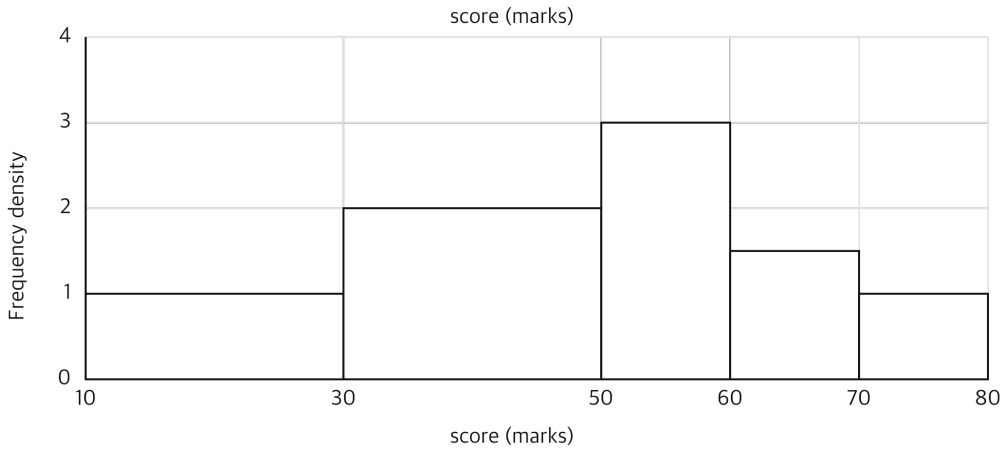
81.8%

Teacher answers & working

Histograms and Frequency Density · Interpreting Histograms

STA-FE00-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



State the modal class.

Answer:

$50 \leq x < 60$

Working:

The modal class is the class with the greatest frequency density.

Look for the tallest bar on the histogram.

The greatest frequency density is 3.

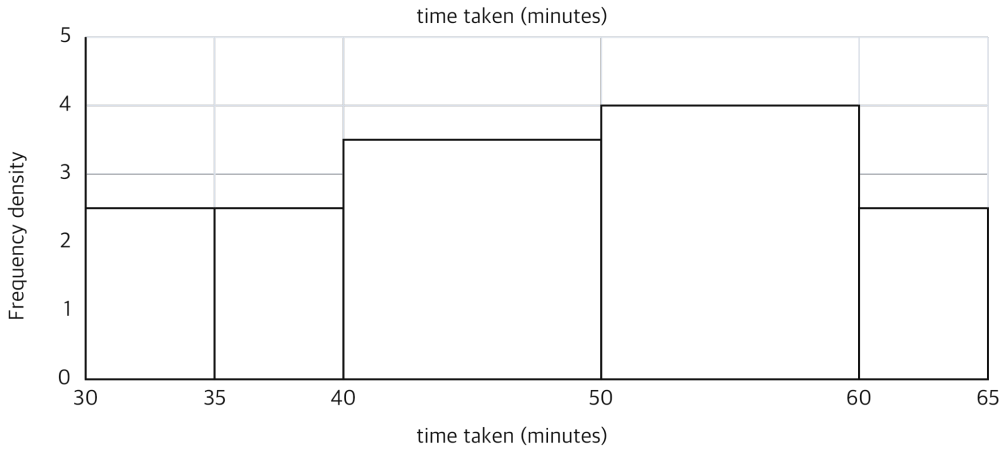
modal class = 50 ≤ x < 60

Teacher answers & working

Histograms and Frequency Density · Interpreting Histograms

STA-FE00-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Find the frequency in the class $40 \leq x < 50$.

Answer:

35 students

Working:

In a histogram, frequency is represented by area.

frequency = class width × frequency density

For $40 \leq x < 50$, class width = 50 - 40 = 10.

The height of the bar is the frequency density, 3.50.

frequency = 10 × 3.50 = 35

35 students